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Editorial Office
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Dear Editors,

Submission: Hardware-Aware Training and Precision-Retention Frontiers for Low-Precision Analog Vision

Article Type: Article

Reviewer Information: Suggested and excluded reviewers are entered through the submission system.

Editorial Summary

We present a behavioural simulation framework for low-precision analog compute-in-memory vision that separates two deployment questions that are often conflated: whether a training objective generalizes across fresh hardware instances, and whether a physical memory substrate remains stable after programming. In an AIHWKit IdealDevice baseline, fixed-instance low-precision analog deployment is stable at 8-bit precision (87.28±0.13% fresh-instance accuracy) but collapses at 4-bit precision (14.64±0.11%). Ensemble Hardware-Aware Training (HAT), which resamples device-to-device mismatch during training, rescues this 4-bit pure-quantization regime and recovers 86.16±0.19% fresh-instance accuracy across three seeds.

We then test whether the algorithmic rescue survives a more realistic phase-change memory (PCM) simulation setting. Under the tested PCM UnitCell regime, 4-bit, 6-bit, and 8-bit models all converge, but retention behaviour becomes precision dependent. The 8-bit PCM model is drift-flat over the tested retention window and provides the strongest overall operating point; the 4-bit PCM model is trainable but incurs a measurable 4.01 percentage-point one-day drift; and the corrected 6-bit PCM model is drift-flat but seed-sensitive, with 68.44% mean fresh-instance accuracy and 6.03 pp seed-to-seed standard deviation. The central message is therefore not simply that HAT improves accuracy, but that distribution-level HAT resolves the cross-instance robustness bottleneck and exposes the remaining physical precision-retention frontier.

Why Nature Electronics?

This work sits at the intersection of analog memory device modelling, mixed-signal system design, and machine-learning robustness. The paper provides a deployment-facing bridge between device-level non-idealities and task-level transformer accuracy, with explicit provenance for the numerical claims and a reproducibility bundle containing the manuscript source, source-data manifests, verification scripts, and canonical JSON evidence. We believe this hardware–algorithm co-design narrative is well aligned with Nature Electronics' readership because it frames low-precision analog deployment as a coupled algorithmic and physical design problem rather than as a pure neural-network training result.

Methodological Rigor

All locked numerical claims are backed by JSON artifacts and guard scripts. The release package includes checks for locked numbers, PCM precision-ladder consistency, bibliography key coverage, DOI/URL resolution, and LaTeX artifact provenance. The manuscript distinguishes canonical claims from diagnostic supplementary analyses and keeps remote cross-architecture and analog-KV-cache results out of the Paper-1 main claim set until their independent gates close.

Companion Work and Collaboration

A separate follow-on line will study analog KV-cache for large language model inference. That work is intentionally kept outside the present manuscript because its evaluation metrics, memory-access pattern, and noise model differ from the vision-training setting reported here.

Key Contributions

1. A behavioural analog-CIM vision workflow that separates cross-instance robustness from physical precision-retention limits.
2. Ensemble HAT, a distribution-matching training protocol that raises 4-bit fresh-instance accuracy from 14.64±0.11% to 86.16±0.19% in the tested pure-quantization regime.
3. A PCM precision ladder showing 8-bit drift-flat behaviour, trainable but drift-limited 4-bit behaviour, and a 6-bit D2D-sensitive transition regime in the tested UnitCell sweep.
4. A reproducibility package with canonical JSON evidence, source-data manifests, and guard scripts for locked-number and reference validation.

Transparency Disclosures

- **Preprint:** None; submitted exclusively to Nature Electronics.
- **Prior Publication:** None; the framework was developed for this submission.
- **Code Availability:** A reviewer-accessible archive accompanies this submission; a public release follows at acceptance.
- **Competing Interests:** The authors declare no competing interests.

The submission package includes a main manuscript and a supplementary information PDF. All datasets are publicly available, and all key claims carry error bars from multiple Monte Carlo runs or training seeds.

Sincerely,

Songqiao Li